

Meta-analysis of early speech and language in children with nonsyndromic cleft palate with or without cleft lip

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Purpose

The literature reports large variation in early speech and language development in young children with cleft palate with or without cleft lip (CP/L). Despite their risk for speech delay secondary to cleft palate, to date no one has completed a meta-analysis to compare the speech skills of young children with CP/L to their non-cleft peers. Roberts, et al. (2012) revealed deficits in language functioning in individuals with cleft when compared to a typical control group; however, Roberts et al.'s (2012) meta-analysis spanned the lifespan, which does not allow us to draw conclusions about language difficulties during the language development period in early childhood and the early school-age years. Therefore, further description of the language abilities of young children with CP/L during the language development period is critically needed.

The purpose of the current project is to characterize the speech and language development of young children with CP/L between birth and age 8 years, 11 months in relation to the development of their non-cleft typical peers. Further, the effects of individual characteristics (e.g., age, cleft type), assessment characteristics (e.g., sampling context, measurement source), and study-level characteristics (e.g., study location, publication year) will be examined for moderating effects.

Research Questions

- To what degree do young children with nonsyndromic CP/L differ from typical peers in speech development according to type of speech analysis (i.e., independent analysis of speech, speech accuracy, and error analysis of speech)?
 - Does age, sampling context, cleft type, study location, or publication year moderate the effect size?
- To what degree do young children with nonsyndromic CP/L differ from typical peers for expressive and receptive language?
 - Does age, sampling context, cleft type, measurement source, study location, or publication year moderate the effect size?

Methods

Data extraction: Studies were coded for study-level characteristics (e.g., publication year and study location) and all qualifying speech and language measures were extracted. For longitudinal studies the last time point was selected. For studies with multiple groups, all groups were entered as a separate effect size. Effect sizes were coded for construct, sub construct, measurement source (standardized test, parent report, language sample), sampling context (single word or language sample; vocabulary or general language skill), and cleft type.

Reliability: Prior to estimating coding reliability, we conducted reliability training on ~40% ($n = 7$) in each domain until coders had 100% reliability. Then coders coded a random 30% selection from remaining articles. Final reliability was 82% for speech and 85% for language.

Analysis: All models were random-effects accounting for within in study dependent effect sizes using robust error variance The robumeta package (Fisher, Tipton, & Zhipeng, 2017) in R was used to analyze the data.

Methods

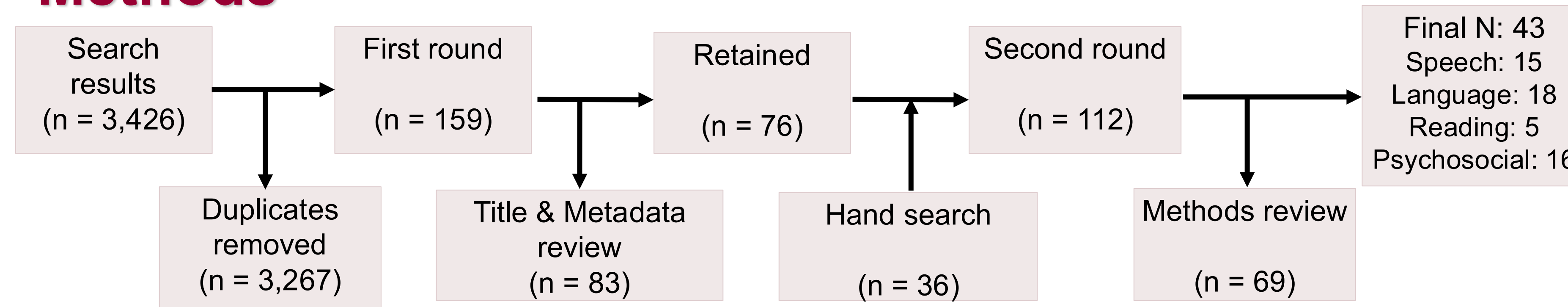


Figure 1. Flow chart of review of search results for inclusion in main project. Only speech and language articles were retained for this presentation.

Search methods: Multiple databases were searched. The inclusion terms were scholarly journal, human participant, “cleft palate.” The exclusion terms were: orthopedics, surgical techniques, and otitis media. Title and metadata were extracted for non-duplicate entries and reviewed for age and appropriate topic. A hand search was completed to ensure all possible articles were included. For Methods review, articles were checked for non-syndromic, non-cleft group, appropriate measures, and statistics for calculating effect sizes.

Design	N	% male	mean	N cleft	Mean age cleft	SD age cleft	Palatal repair	N TD	Mean age TD	SD age TD	N USA	N 2006 <
Speech												
Group	7	53.67		158	42.41	20.69	[3.4, 12.5]	171	42.47	21.40	4	4
Intervention	1	45		10	27.40	.	12	10	20.20		1	1
Longitudinal	7	60.09		136	41.37	29.33	[6.55, 14.3]	99	40.79	30.39	4	4
Language												
Group	10	56.08		482	51.59	26.22	[4.6, 13.3]	482	52.26	25.88	6	4
Intervention	1	45		10	27.40	.	12	10	20.20	.	1	1
Longitudinal	7	60.99		192	25.22	6.25	[11.1, 13.3]	174	25.11	6.44	6	3

Table 1. Descriptive information about the studies included for early speech and language analysis. Notes. SD = Standard deviation. TD = noncleft typical developing peers

Results

Table 2. Metaregression results for speech and language sub constructs.

Measure type	<i>n</i>	<i>k</i>	<i>ES_d</i>	<i>SE</i>	<i>df</i>	<i>p</i>	95% CI	τ^2
Speech								
Independent	7	24	-1.57	0.39	5.94	0.01	[-2.51, -0.62]	1.19
Accuracy	10	43	-1.12	0.32	8.97	0.01	[-1.85, -0.40]	1.58
Error	5	35	1.29	0.36	3.99 ^a	0.02	[0.27, 2.30]	1.16
Language								
Expressive language	16	59	-0.53	0.08	13.6	0.00	[-0.71, -0.36]	0.09
Receptive language	11	17	-0.63	0.13	9.39	0.00	[-0.91, -0.34]	0.14

Note. ^aIf *df* are < 4 interpret results with caution and do not run moderators. Effect sizes were calculated as Cleft – Control.

Results

Table 3. Moderator analysis for sub constructs of speech.

Variable	Speech									
	Independent					Accuracy				
	<i>B</i>	<i>SE</i>	<i>df</i>	<i>p</i>	95% CI	<i>B</i>	<i>SE</i>	<i>df</i>	<i>p</i>	95% CI
Age	-0.08	0.06	3.61	0.26	[-0.27, 0.02]	-0.01	0.01	4.19	0.82	[-0.03, 0.03]
Sampling context	-0.01	1.15	1.97	0.99	[-5.02, 5.00]	0.76	0.61	8.24	0.25	[-0.65, 2.16]
Cleft type	-0.35	0.86	1.50	0.73	[-4.82, 5.52]	0.02	0.34	7.98	0.95	[-0.75, 0.80]
Location	0.25	0.60	1.57	0.73	[-3.16, 3.65]	0.30	0.55	6.50	0.59	[-1.01, 1.62]
Publication Year	-1.75	0.40	3.97	0.01	[-2.88, -0.63]	-0.13	0.47	3.71	0.79	[-1.47, 1.21]

Table 4. Moderator analysis for sub constructs of language.

Variable	Language									
	Expressive					Receptive				
	<i>B</i>	<i>SE</i>	<i>df</i>	<i>p</i>	95% CI	<i>B</i>	<i>SE</i>	<i>df</i>	<i>p</i>	95% CI
Age	0.01	0.00	5.43	0.01	[0.00, 0.01]	-0.00	0.01	5.16	0.95	[-0.01, 0.01]
Sampling context	-0.21	0.12	9.90	0.12	[-0.48, 0.06]	-0.41	0.34	3.06	0.31	[-1.46, 0.65]
Cleft type	-0.01	0.09	5.31	0.88	[-0.25, 0.22]	0.02	0.11	2.70	0.86	[-0.37, 0.41]
Measure source	-0.24	0.12	8.69	0.07	[-0.51, 0.03]	-	-	-	-	-
Location	0.18	0.15	3.90	0.32	[-0.26, 0.61]	0.09	0.14	8.61	0.52	[-0.23, 0.42]
Publication Year	0.18	0.15	11.54	0.27	[-0.16, 0.51]	0.39	0.22	6.13	0.13	[-0.15, 0.94]

Conclusion

Speech: Young children with CP/L demonstrated significantly lower speech skills and significantly more speech errors. Individual and assessment characteristics did not moderate the mean effect sizes. Publication year moderated effect sizes for independent speech analysis, indicating that studies after 2006 have reported larger effect sizes. The significantly different performance of children with CP/L in comparison to non-cleft peers across speech analysis types suggests that clinicians should utilize multiple methods of speech analysis when conducting speech evaluations with young children with CP/L.

Language: Young children with CP/L demonstrated significantly lower expressive and receptive language skills than their typical peers. Age moderated expressive language indicating that differences decreased with age. The significant difference for receptive language was not explained by individual, assessment, or study characteristics, which may indicate that receptive language is not improving alongside expressive language.

For more information, visit:

<https://tinyurl.com/yb96pbmf>